Readme

1. This Appendix presents the validation process with Laplace's equation for second- and third-order potential gradients in Tables 4-7 in Appendix A and 1-2.
2. Each of the following sections is independent by using the command **Clear["Global`*"]** to clear previous arguments.
3. After running all codes, the results of Laplace's equation are obtained as zero.

1. Table 4 in Appendix A

In[1]:=
```
Clear["Global`*"];
```

In[2]:=
```
l = Sqrt[r^2 + r3^2 - 2 r r3 Cos[θ3]] ; A = l + r3 - r Cos[θ3]; U = Sqrt[r^2 + r3^2 - 2 r r3 Cos[θ3]] / Sqrt[r^2 + r3^2] ;
```

In[3]:=
```
VxxP = G ρ / 48
  (- 3 Cos[θ3] (8 (λ2 - λ1) Sin[θ3]^2 + 4 Cos[2 λ - λ1 - λ2] * Sin[λ2 - λ1] (5 - Cos[2 θ3]))
    Log[A] + 1 / (r^3 l) (-4 (r^4 + r^2 r3^2 + 4 r3^4) (λ2 - λ1) + 2 r^2 ((3 r^2 + r3^2) Cos[2 θ3] -
        3 r r3 Cos[3 θ3]) (2 λ2 - 2 λ1 + Sin[2 (λ - λ1)] - Sin[2 (λ - λ2)]) -
      4 (13 r^4 + 13 r^2 r3^2 + 4 r3^4) Cos[2 λ - λ1 - λ2] * Sin[λ2 - λ1] + 4 r r3 Cos[θ3]
       ((r^2 + 4 r3^2) (λ2 - λ1) + (25 r^2 + 4 r3^2) Cos[2 λ - λ1 - λ2] * Sin[λ2 - λ1])));
```

In[4]:=
```
VyyP = G ρ / 48
  (- 3 Cos[θ3] (8 (λ2 - λ1) Sin[θ3]^2 - 4 Cos[2 λ - λ1 - λ2] * Sin[λ2 - λ1] (5 - Cos[2 θ3]))
    Log[A] + 1 / (r^3 l) (-4 (r^4 + r^2 r3^2 + 4 r3^4) (λ2 - λ1) + 2 r^2 ((3 r^2 + r3^2) Cos[2 θ3] -
        3 r r3 Cos[3 θ3]) (2 λ2 - 2 λ1 - Sin[2 (λ - λ1)] + Sin[2 (λ - λ2)]) +
      4 (13 r^4 + 13 r^2 r3^2 + 4 r3^4) Cos[2 λ - λ1 - λ2] * Sin[λ2 - λ1] + 4 r r3 Cos[θ3]
       ((r^2 + 4 r3^2) (λ2 - λ1) - (25 r^2 + 4 r3^2) Cos[2 λ - λ1 - λ2] * Sin[λ2 - λ1])));
```

In[5]:=
```
VzzP =
  G ρ (λ2 - λ1) / (6 r^3) (1 / l (r^4 + r^2 r3^2 + 4 r3^4 - r r3 (r^2 + 4 r3^2) Cos[θ3] - r^2 (3 r^2 + r3^2) Cos[2 θ3] +
      3 r^3 r3 Cos[3 θ3]) + 6 r^3 Sin[θ3]^2 Cos[θ3] * Log[A]);
```

In[6]:=
$$\text{VxxzP} = \frac{G * \rho}{16 * r^4 * l^3}$$
$$\left(6 * r^2 * r3^4 * \text{Cos}[2 * \theta 3] * (2 * \lambda 2 - 2 * \lambda 1 + \text{Sin}[2 * (\lambda - \lambda 1)] - \text{Sin}[2 * (\lambda - \lambda 2)]) - \right.$$
$$\text{Cos}[2 * \lambda - \lambda 1 - \lambda 2] * \text{Sin}[\lambda 2 - \lambda 1] *$$
$$\left(5 * r^6 * U + r3^6 * (5 * U - 16) + r^2 * r3^4 * (15 * U - 52) + r^4 * r3^2 * (15 * U - 16)\right) + 2 * r *$$
$$r3 * \text{Cos}[\theta 3] * \left((\lambda 2 - \lambda 1) * \left(5 * r^4 * U + r3^4 * (5 * U - 24) + 2 * r^2 * r3^2 * (5 * U - 4)\right) + \right.$$
$$\left.\left(r^2 + r3^2\right) * \text{Cos}[2 * \lambda - \lambda 1 - \lambda 2] * \text{Sin}[\lambda 2 - \lambda 1] * \left(5 * r^2 * U + r3^2 * (5 * U - 24)\right)\right) -$$
$$\left.5 \left(r^2 + r3^2\right)^3 * (\lambda 2 - \lambda 1) * U + 4 * r3^4 * \left(9 * r^2 + 4 * r3^2\right) * (\lambda 2 - \lambda 1)\right);$$

In[7]:=
$$\text{VyyzP} = \frac{G * \rho}{16 * r^4 * l^3}$$
$$\left(6 * r^2 * r3^4 * \text{Cos}[2 * \theta 3] * (2 * \lambda 2 - 2 * \lambda 1 - \text{Sin}[2 * (\lambda - \lambda 1)] + \text{Sin}[2 * (\lambda - \lambda 2)]) + \right.$$
$$\text{Cos}[2 * \lambda - \lambda 1 - \lambda 2] * \text{Sin}[\lambda 2 - \lambda 1] *$$
$$\left(5 * r^6 * U + r3^6 * (5 * U - 16) + r^2 * r3^4 * (15 * U - 52) + r^4 * r3^2 * (15 * U - 16)\right) + 2 * r *$$
$$r3 * \text{Cos}[\theta 3] * \left((\lambda 2 - \lambda 1) * \left(5 * r^4 * U + r3^4 * (5 * U - 24) + 2 * r^2 * r3^2 * (5 * U - 4)\right) - \right.$$
$$\left.\left(r^2 + r3^2\right) * \text{Cos}[2 * \lambda - \lambda 1 - \lambda 2] * \text{Sin}[\lambda 2 - \lambda 1] * \left(5 * r^2 * U + r3^2 * (5 * U - 24)\right)\right) -$$
$$\left.5 \left(r^2 + r3^2\right)^3 * (\lambda 2 - \lambda 1) * U + 4 * r3^4 * \left(9 * r^2 + 4 * r3^2\right) * (\lambda 2 - \lambda 1)\right);$$

In[8]:=
$$\text{VzzzP} = -\frac{G * \rho * (\lambda 2 - \lambda 1)}{2 * r^4 * l^3}$$
$$\left(r3^3 * \left(9 * r^2 * r3 + 4 * r3^3 - 4 * r * \left(r^2 + 3 * r3^2\right) * \text{Cos}[\theta 3] + 3 * r^2 * r3 * \text{Cos}[2 * \theta 3]\right)\right);$$

In[9]:=
```
FullSimplify[VxxP + VyyP + VzzP]
```

Out[9]=
```
0
```

In[10]:=
```
FullSimplify[VxxzP + VyyzP + VzzzP]
```

Out[10]=
$$\frac{5\, G \left(r^2 + r3^2\right)^{3/2} (\lambda 1 - \lambda 2)\, \rho}{8\, r^4}$$

In[11]:=
$$f[\theta 3_] := \frac{5\, G \left(r^2 + r3^2\right)^{3/2} (\lambda 1 - \lambda 2)\, \rho}{8\, r^4};\ f[\theta 2] - f[\theta 1]$$

Out[11]=
```
0
```

2. Table 5 in Appendix A

In[12]:=
```
Clear["Global`*"];
```

In[13]:=
$$l = \sqrt{r^2 + r3^2 - 2\, r\, r3\, \text{Cos}[\theta 3]};\ A = l + r3 - r\, \text{Cos}[\theta 3];\ U = \frac{\sqrt{r^2 + r3^2 - 2\, r\, r3\, \text{Cos}[\theta 3]}}{\sqrt{r^2 + r3^2}};$$

In[14]:=
$$\text{VxxZB} = -\frac{\text{Pi}\, G\, \rho}{6\, r^3} \left(\frac{1}{l}\left(r^4 + r^2\, r3^2 + 4\, r3^4 - r\, r3\, \left(r^2 + 4\, r3^2\right) \text{Cos}[\theta 3] - \right.\right.$$
$$\left.\left.r^2 \left(3\, r^2 + r3^2\right) \text{Cos}[2\, \theta 3] + 3\, r^3\, r3\, \text{Cos}[3\, \theta 3]\right) + 6\, r^3\, \text{Sin}[\theta 3]^2 * \text{Cos}[\theta 3] * \text{Log}[A]\right);$$

In[15]:=
$$\text{VyyZB} = -\frac{\text{Pi G}\,\rho}{6\,r^3}\left(\frac{1}{l}\left(r^4 + r^2\,r3^2 + 4\,r3^4 - r\,r3\left(r^2 + 4\,r3^2\right)\text{Cos}[\theta 3] - r^2\left(3\,r^2 + r3^2\right)\text{Cos}[2\,\theta 3] + 3\,r^3\,r3\,\text{Cos}[3\,\theta 3]\right) + 6\,r^3\,\text{Sin}[\theta 3]^2 * \text{Cos}[\theta 3] * \text{Log}[A]\right);$$

In[16]:=
$$\text{VzzZB} = \frac{\text{Pi G}\,\rho}{3\,r^3}\left(\frac{1}{l}\left(r^4 + r^2\,r3^2 + 4\,r3^4 - r\,r3\left(r^2 + 4\,r3^2\right)\text{Cos}[\theta 3] - r^2\left(3\,r^2 + r3^2\right)\text{Cos}[2\,\theta 3] + 3\,r^3\,r3\,\text{Cos}[3\,\theta 3]\right) + 6\,r^3\,\text{Sin}[\theta 3]^2 * \text{Cos}[\theta 3] * \text{Log}[A]\right);$$

In[17]:=
$$\text{VxxzZB} = \frac{\text{Pi G}\,\rho}{8\,r^4\,l^3}\left(4 * r3^4 * \left(9 * r^2 + 4 * r3^2 + 3 * r^2 * \text{Cos}[2\,\theta 3]\right) + 2 * r * r3 * \text{Cos}[\theta 3] * \left(5 * r^4 * U + r3^4 * (5 * U - 24) + 2 * r^2 * r3^2 * (5 * U - 4)\right) - 5 * \left(r^2 + r3^2\right)^3 * U\right);$$

In[18]:=
$$\text{VyyzZB} = \frac{\text{Pi G}\,\rho}{8\,r^4\,l^3}\left(4 * r3^4 * \left(9 * r^2 + 4 * r3^2 + 3 * r^2 * \text{Cos}[2\,\theta 3]\right) + 2 * r * r3 * \text{Cos}[\theta 3] * \left(5 * r^4 * U + r3^4 * (5 * U - 24) + 2 * r^2 * r3^2 * (5 * U - 4)\right) - 5 * \left(r^2 + r3^2\right)^3 * U\right);$$

In[19]:=
$$\text{VzzzZB} = -\frac{\text{Pi G}\,\rho\,r3^3\left(9\,r^2\,r3 + 4\,r3^3 - 4\,r\left(r^2 + 3\,r3^2\right)\text{Cos}[\theta 3] + 3\,r^2\,r3\,\text{Cos}[2\,\theta 3]\right)}{r^4\,l^3};$$

In[20]:=
```
FullSimplify[VxxZB + VyyZB + VzzZB]
```

Out[20]=
$$0$$

In[21]:=
```
FullSimplify[VxxzZB + VyyzZB + VzzzZB]
```

Out[21]=
$$-\frac{5\,G\,\pi\left(r^2 + r3^2\right)^{3/2}\rho}{4\,r^4}$$

In[22]:=
$$g[\theta 3_] := -\frac{5\,G\,\pi\left(r^2 + r3^2\right)^{3/2}\rho}{4\,r^4};\ g[\theta 2] - g[\theta 1]$$

Out[22]=
$$0$$

3. Table 6 in Appendix A

In[23]:=
```
Clear["Global`*"];
```

In[24]:=
$$l1 = \sqrt{r^2 + r1^2 - 2\,r\,r1\,\text{Cos}[\theta 3]};\ l2 = \sqrt{r^2 + r2^2 - 2\,r\,r2\,\text{Cos}[\theta 3]};\ B = \frac{l1 + l2 + (r2 - r1)}{l1 + l2 - (r2 - r1)};$$

In[25]:=
$$\text{VxxP} = \frac{G\rho}{48}\left(-3\,\text{Cos}[\theta 3]\,\left(8\,(\lambda 2-\lambda 1)\,\text{Sin}[\theta 3]^2+4\,\text{Cos}[2\lambda-\lambda 1-\lambda 2]*\text{Sin}[\lambda 2-\lambda 1]\,(5-\text{Cos}[2\theta 3])\right)\right.$$
$$\text{Log}[B]+\frac{1}{r^3\,l}\left(-4\left(r^4+r^2\,r3^2+4\,r3^4\right)(\lambda 2-\lambda 1)+2\,r^2\left(\left(3\,r^2+r3^2\right)\text{Cos}[2\theta 3]-\right.\right.$$
$$3\,r\,r3\,\text{Cos}[3\theta 3]\Big)\,(2\lambda 2-2\lambda 1+\text{Sin}[2\,(\lambda-\lambda 1)]-\text{Sin}[2\,(\lambda-\lambda 2)])-$$
$$4\left(13\,r^4+13\,r^2\,r3^2+4\,r3^4\right)\text{Cos}[2\lambda-\lambda 1-\lambda 2]*\text{Sin}[\lambda 2-\lambda 1]+4\,r\,r3\,\text{Cos}[\theta 3]$$
$$\left.\left.\left(\left(r^2+4\,r3^2\right)(\lambda 2-\lambda 1)+\left(25\,r^2+4\,r3^2\right)\text{Cos}[2\lambda-\lambda 1-\lambda 2]*\text{Sin}[\lambda 2-\lambda 1]\right)\right)\right);$$

In[26]:=
$$\text{VyyP} = \frac{G\rho}{48}\left(-3\,\text{Cos}[\theta 3]\,\left(8\,(\lambda 2-\lambda 1)\,\text{Sin}[\theta 3]^2-4\,\text{Cos}[2\lambda-\lambda 1-\lambda 2]*\text{Sin}[\lambda 2-\lambda 1]\,(5-\text{Cos}[2\theta 3])\right)\right.$$
$$\text{Log}[B]+\frac{1}{r^3\,l}\left(-4\left(r^4+r^2\,r3^2+4\,r3^4\right)(\lambda 2-\lambda 1)+2\,r^2\left(\left(3\,r^2+r3^2\right)\text{Cos}[2\theta 3]-\right.\right.$$
$$3\,r\,r3\,\text{Cos}[3\theta 3]\Big)\,(2\lambda 2-2\lambda 1-\text{Sin}[2\,(\lambda-\lambda 1)]+\text{Sin}[2\,(\lambda-\lambda 2)])+$$
$$4\left(13\,r^4+13\,r^2\,r3^2+4\,r3^4\right)\text{Cos}[2\lambda-\lambda 1-\lambda 2]*\text{Sin}[\lambda 2-\lambda 1]+4\,r\,r3\,\text{Cos}[\theta 3]$$
$$\left.\left.\left(\left(r^2+4\,r3^2\right)(\lambda 2-\lambda 1)-\left(25\,r^2+4\,r3^2\right)\text{Cos}[2\lambda-\lambda 1-\lambda 2]*\text{Sin}[\lambda 2-\lambda 1]\right)\right)\right);$$

In[27]:=
$$\text{VzzP} = \frac{G\rho\,(\lambda 2-\lambda 1)}{6\,r^3}\left(\frac{1}{l}\left(r^4+r^2\,r3^2+4\,r3^4-r\,r3\left(r^2+4\,r3^2\right)\text{Cos}[\theta 3]-r^2\left(3\,r^2+r3^2\right)\text{Cos}[2\theta 3]+\right.\right.$$
$$\left.\left.3\,r^3\,r3\,\text{Cos}[3\theta 3]\right)+6\,r^3\,\text{Sin}[\theta 3]^2\,\text{Cos}[\theta 3]*\text{Log}[B]\right);$$

In[28]:=
```
FullSimplify[VxxP + VyyP + VzzP]
```
Out[28]=
```
0
```

4. Table 7 in Appendix A

In[29]:=
```
Clear["Global`*"];
```

In[30]:=
$$l = \sqrt{r^2+r3^2-2\,r\,r3\,\text{Cos}[\theta 3]};$$
$$l1 = \sqrt{r^2+r1^2-2\,r\,r1\,\text{Cos}[\theta 3]};$$
$$l2 = \sqrt{r^2+r2^2-2\,r\,r2\,\text{Cos}[\theta 3]};$$
$$B = \frac{l1+l2+(r2-r1)}{l1+l2-(r2-r1)};$$

In[31]:=
$$\text{VxxZB} = -\frac{\text{Pi}\,G\rho}{6\,r^3}\left(\frac{1}{l}\left(r^4+r^2\,r3^2+4\,r3^4-r\,r3\left(r^2+4\,r3^2\right)\text{Cos}[\theta 3]-\right.\right.$$
$$\left.\left.r^2\left(3\,r^2+r3^2\right)\text{Cos}[2\theta 3]+3\,r^3\,r3\,\text{Cos}[3\theta 3]\right)+6\,r^3\,\text{Sin}[\theta 3]^2*\text{Cos}[\theta 3]*\text{Log}[B]\right);$$

In[32]:=
$$\text{VyyZB} = -\frac{\text{Pi G}\,\rho}{6\,r^3}\left(\frac{1}{l}\left(r^4 + r^2\,r3^2 + 4\,r3^4 - r\,r3\left(r^2 + 4\,r3^2\right)\text{Cos}[\theta 3] - r^2\left(3\,r^2 + r3^2\right)\text{Cos}[2\,\theta 3] + 3\,r^3\,r3\,\text{Cos}[3\,\theta 3]\right) + 6\,r^3\,\text{Sin}[\theta 3]^2 * \text{Cos}[\theta 3] * \text{Log}[B]\right);$$

In[33]:=
$$\text{VzzZB} = \frac{\text{Pi G}\,\rho}{3\,r^3}\left(\frac{1}{l}\left(r^4 + r^2\,r3^2 + 4\,r3^4 - r\,r3\left(r^2 + 4\,r3^2\right)\text{Cos}[\theta 3] - r^2\left(3\,r^2 + r3^2\right)\text{Cos}[2\,\theta 3] + 3\,r^3\,r3\,\text{Cos}[3\,\theta 3]\right) + 6\,r^3\,\text{Sin}[\theta 3]^2 * \text{Cos}[\theta 3] * \text{Log}[B]\right);$$

In[34]:=
```
FullSimplify[VxxZB + VyyZB + VzzZB]
```

Out[34]=
```
0
```

5. Table 1

In[35]:=
```
Clear["Global`*"];
```

In[36]:=
$$\text{VxxLS} = \text{G}\,\rho\,\text{Cos}[2\,\lambda - \lambda 1 - \lambda 2]\,\text{Sin}[\lambda 2 - \lambda 1]\left(\text{Log}\left[\frac{r - r1}{r - r2}\right] + \text{Log}\left[\frac{r + r2}{r + r1}\right]\right) - \frac{2\,\text{G}\,\rho}{3\,r^3}\left(r3^3\,(\lambda 2 - \lambda 1) + r3\,\text{Cos}[2\,\lambda - \lambda 1 - \lambda 2]\,\text{Sin}[\lambda 2 - \lambda 1]\left(3\,r^2 + r3^2\right)\right);$$

In[37]:=
$$\text{VyyLS} = -\text{G}\,\rho\,\text{Cos}[2\,\lambda - \lambda 1 - \lambda 2]\,\text{Sin}[\lambda 2 - \lambda 1]\left(\text{Log}\left[\frac{r - r1}{r - r2}\right] + \text{Log}\left[\frac{r + r2}{r + r1}\right]\right) - \frac{2\,\text{G}\,\rho}{3\,r^3}\left(r3^3\,(\lambda 2 - \lambda 1) - r3\,\text{Cos}[2\,\lambda - \lambda 1 - \lambda 2]\,\text{Sin}[\lambda 2 - \lambda 1]\left(3\,r^2 + r3^2\right)\right);$$

In[38]:=
$$\text{VzzLS} = \frac{4\,\text{G}\,\rho\,(\lambda 2 - \lambda 1)}{3\,r^3}\,r3^3;$$

In[39]:=
$$\text{VxxzLS} = \frac{2\,\text{G}\,\rho\,r2^3}{r^4}\left((\lambda 2 - \lambda 1) - \frac{r2^2\,\text{Cos}[2\,\lambda - \lambda 1 - \lambda 2] * \text{Sin}[\lambda 2 - \lambda 1]}{r^2 - r2^2}\right) - \frac{2\,\text{G}\,\rho\,r1^3}{r^4}\left((\lambda 2 - \lambda 1) - \frac{r1^2\,\text{Cos}[2\,\lambda - \lambda 1 - \lambda 2] * \text{Sin}[\lambda 2 - \lambda 1]}{r^2 - r1^2}\right);$$

In[40]:=
$$\text{VyyzLS} = \frac{2\,\text{G}\,\rho\,r2^3}{r^4}\left((\lambda 2 - \lambda 1) + \frac{r2^2\,\text{Cos}[2\,\lambda - \lambda 1 - \lambda 2] * \text{Sin}[\lambda 2 - \lambda 1]}{r^2 - r2^2}\right) - \frac{2\,\text{G}\,\rho\,r1^3}{r^4}\left((\lambda 2 - \lambda 1) + \frac{r1^2\,\text{Cos}[2\,\lambda - \lambda 1 - \lambda 2] * \text{Sin}[\lambda 2 - \lambda 1]}{r^2 - r1^2}\right);$$

In[41]:=
$$\text{VzzzLS} = -\frac{4\,\text{G}\,\rho\,(\lambda 2 - \lambda 1)}{r^4}\left(r2^3 - r1^3\right);$$

In[42]:=
```
FullSimplify[VxxLS + VyyLS + VzzLS]
```

Out[42]=
```
0
```

In[43]:=
```
FullSimplify[VxxzLS + VyyzLS + VzzzLS]
```

Out[43]=
```
0
```

6. Table 2

In[44]:=
```
Clear["Global`*"];
```

In[45]:=
$$\text{VxxSH} = -\frac{4\,\text{Pi}\,\text{G}\,\rho\,\left(\text{r2}^3 - \text{r1}^3\right)}{3\,\text{r}^3};$$

In[46]:=
$$\text{VyySH} = -\frac{4\,\text{Pi}\,\text{G}\,\rho\,\left(\text{r2}^3 - \text{r1}^3\right)}{3\,\text{r}^3};$$

In[47]:=
$$\text{VzzSH} = \frac{8\,\text{Pi}\,\text{G}\,\rho\,\left(\text{r2}^3 - \text{r1}^3\right)}{3\,\text{r}^3};$$

In[48]:=
$$\text{VxxzSH} = \frac{4\,\text{Pi}\,\text{G}\,\rho\,\left(\text{r2}^3 - \text{r1}^3\right)}{\text{r}^4};$$

In[49]:=
$$\text{VyyzSH} = \frac{4\,\text{Pi}\,\text{G}\,\rho\,\left(\text{r2}^3 - \text{r1}^3\right)}{\text{r}^4};$$

In[50]:=
$$\text{VzzzSH} = -\frac{8\,\text{Pi}\,\text{G}\,\rho\,\left(\text{r2}^3 - \text{r1}^3\right)}{\text{r}^4};$$

In[51]:=
```
FullSimplify[VxxSH + VyySH + VzzSH]
```

Out[51]=
```
0
```

In[52]:=
```
FullSimplify[VxxzSH + VyyzSH + VzzzSH]
```

Out[52]=
```
0
```